Candidate Name: ______________________________________ Class: 07S________

JURONG JUNIOR COLLEGE

PRELIMINARY EXAMINATION 2008

H1 BIOLOGY 8875/2

FRIDAY 15 AUGUST 2008 2 HOURS

INSTRUCTIONS TO CANDIDATES :

Write your name and class in the spaces at the top of this page and on all the work you hand in.
Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.
Write your answer in the spaces provided on the question paper.

Section B
Answer any **one** question.

At the end of the examination
1. Fasten any separate answer paper used securely to the question paper.
2. Enter the question number from Section B in the grid opposite.

EXAMINER'S USE ONLY	
Section A	**Mark**
1	
2	
3	
4	
Section B	
Total	

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

Section A

Answer **all** the questions in this section.

1. During digestion, large molecules are broken down into their constituent monomers (subunits).

(a) Complete the table below. [2]

Large molecules	**Monomers (subunits)**
	Amino acids
Lipid	
	Monosaccharide
DNA	

(b) Explain how a monosaccharide enters an epithelial cell. [2]

Starch, a storage molecule in plants, is a polymer made of glucose molecules joined together. It consists of two components.

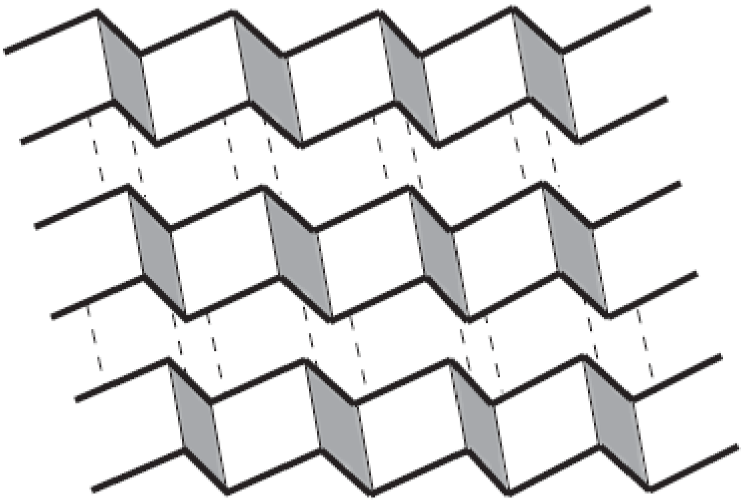
(c) Name these two components. [2]

(d) Give one similarity and one difference between these two components. [2]

Similarity

Difference

Protein molecules come in many shapes and forms that can be classified into primary, secondary, tertiary and quaternary. The secondary, tertiary and quaternary shapes arise as a result of different kinds of folding of a primary structure. One kind of secondary structure is a pleated sheet where the primary molecule extends along the folded sheet. The primary structures in the layers are held together by hydrogen bonding.



(e) Suggest two ways in which such a structure may be important to the functioning of the protein. [2]

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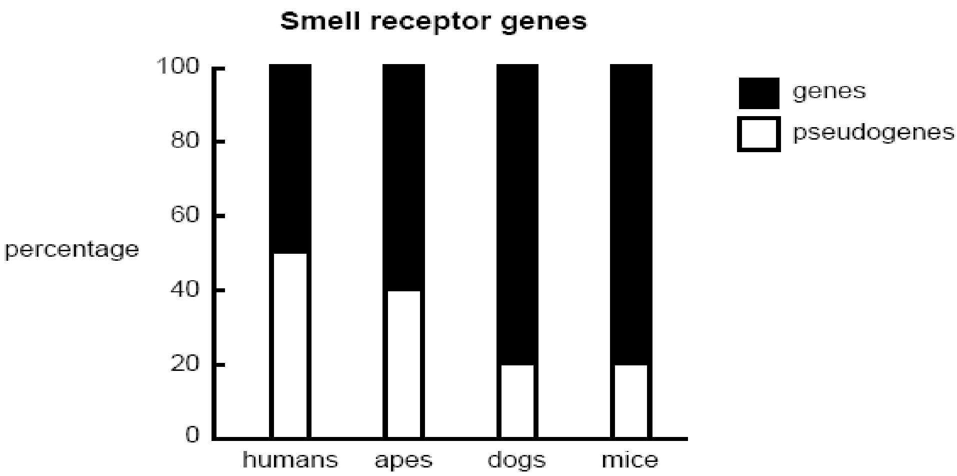
[Total: 10]

2. Pseudogenes are the remains of broken genes which are unable to function and can be considered to be genetic fossils. Some are relics of genes lost through evolution while others reflect an earlier version of a present functional gene. Pseudogenes are able to accumulate all kinds of mutations.

   In mammals, more than one thousand genes coding for smell receptors have been identified. Individual species vary in the proportion of these genes that have become pseudogenes, and humans have fewer than 500 functional genes coding for smell receptors. The majority of these genes are still functional genes in rats and mice.



(a)

(i) From the graph, what can be concluded about the importance of smell to the survival of these four groups? [2]

(ii) Mice and dogs can only distinguish shades of grey, while some apes and monkeys are able to distinguish colours. Would you expect to find any change in the numbers of smell pseudogenes in apes and monkeys able to distinguish colours? Explain why. [2]

(iii) Why are pseudogenes able to accumulate mutations that do not exist in functional genes? [2]

Body cells have elaborate control systems that regulate their division. If these control systems break down, cells start to grow and divide in an unregulated way and produce a tumour.

(b) Cigarette smoking can lead to the formation of tumours. Describe how. [4]

[Total: 10]

3. Cytochrome c is found in the mitochondria of every aerobic eukaryote — animal, plant, and fungi. The amino acid sequences of cytochrome c from many of these organisms have been determined, and comparison made between the molecules shows that they are related.

Human cytochrome c contains 104 amino acids, and 37 of these have been found at equivalent positions in every cytochrome c that has been sequenced. **These molecules are homologous**.

Table 8 shows the first 18 amino acid residues of human cytochrome c, from the N-terminal, with the corresponding sequences from six other organisms aligned beneath. A ditto (“) indicates that the amino acid is the same one found at that position in the human molecule. All the vertebrate cytochromes (the first four) start with glycine (Gly). The *Drosophila*, wheat, and yeast cytochromes have several amino acids that precede the sequence shown here (indicated by <<<).

Cytochrome	Amino acid																		
		1					6				10				14			17	18
Human		Gly	Asp	Val	Glu	Lys	Gly	Lys	Lys	Ile	Phe	Ile	Met	Lys	Cys	Ser	Gln	Cys	His
Pig		“	“	“	“	“	“	“	“	“	“	Val	Gln	“	“	Ala	“	“	“
Chicken		“	“	Ile	“	“	“	“	“	“	“	Val	Gln	“	“	“	“	“	“
Dogfish		“	“	“	“	“	“	“	“	Val	“	Val	Gln	“	“	Ala	“	“	“
Drosophila	<<<	“	“	“	“	“	“	“	“	Leu	“	Val	Gln	Arg	“	Ala	“	“	“
Wheat	<<<	“	Asn	Pro	Asp	Ala	“	Ala	“	“	“	Lys	Thr	“	“	Ala	“	“	“
Yeast	<<<	“	Ser	Ala	Lys	“	“	Ala	Thr	Leu	“	Lys	Thr	Arg	“	“	“	“	“

(a) With reference to the table above, explain what is meant by “These molecules are homologous”. [3]

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(b) (i) Explain why the analysis of amino acid sequence is used to establish evolutionary relationship. [3]

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(ii) State a limitation of using difference in amino acid sequences of only one protein (cytochrome c) to establish phylogenetic relationship among the organisms. [1]

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Cytochrome c is an ancient molecule which has coding and non-coding regions. It has evolved very slowly. Even after more than 2 billion years, one-third of its amino acids are unchanged. The conserved sequences are a great help in working out the evolutionary relationships between distantly-related organisms like fish and humans. However, for humans and the apes (e.g. chimpanzee), their cytochrome c molecules are identical and evolutionary relationships cannot be determined.

(c) State the name given to

Coding regions ___________________

Non-coding regions __________________ [2]

(d) Suggest an alternative way by which the evolutionary relationship between humans and the apes can be determined by molecular techniques. [1]

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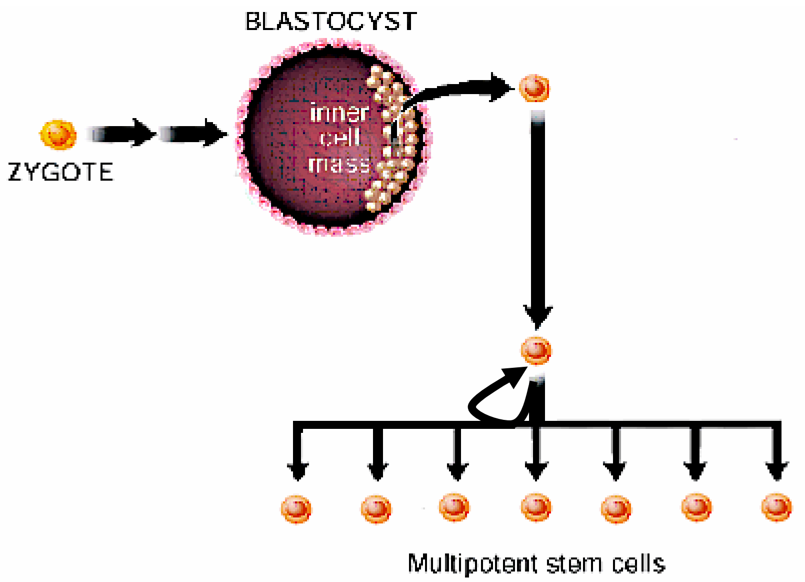
[Total: 10]

4. The figure below shows how a one-celled zygote undergoes various cellular divisions to give rise to other cell types.



(a) Describe two unique features of stem cells that could be seen from the figure above. [2]

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(b) Describe the developmental potential of the one-celled zygote and the inner cell mass of the blastocyst. [4]

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(c) Multipotent stem cells can form specialized cell types of the tissue in which they reside. Give an example of a multipotent stem cell and the specialized cell types that it can formed. [2]

(d) Adult stem cells are multipotent stem cells found among differentiated cells in a tissue or organ. What are the primary roles of these adult stem cells and describe one appropriate unique feature of this type of stem cells that is not mentioned in (a). [2]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
You answers must be in continuous prose, where appropriate.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

5.
(a) Describe the structure of DNA and its arrangement in a chromosome. [8]

(b) Compare and contrast between DNA replication and transcription. [12]

OR

6.
(a) Describe the polymerase chain reaction and explain the advantages and limitations of this procedure. [12]

(b) Discuss the goals and implications of the human genome project, including the benefits and ethical concerns. [8]